

WORK INTEGRATED LEARNING, OCTOBER 2026

Toolshed Hire, Task 2

The built system for INSY7315

- Teon Kleynhans, ST10434209, working alone
- Live at toolshed-hire.vercel.app



TOOLSHED
HIRE

Rental management for a tool hire business
with three branches in Cape Town

The problem

3

branches

400

tagged units

120

models



Bookings live in a paper diary and a WhatsApp group

A tool hire business with three branches in Cape Town

1

The same unit is promised twice

Double booking, which the database itself now prevents

2

Staff phone each other to find stock

One search now answers for all three branches

3

The owner cannot say which equipment earns its keep

The reporting that answers it comes in the final release



What Task 2 delivers



Customer journey

From the catalogue to a confirmed booking



Counter journey

From a walk-in to checkout, return and deposit settlement



Late fees and deposits

One late fee policy, and the deposit settled at the counter



Two environments

Staging and production, deployed by the pipeline only

18 of 24

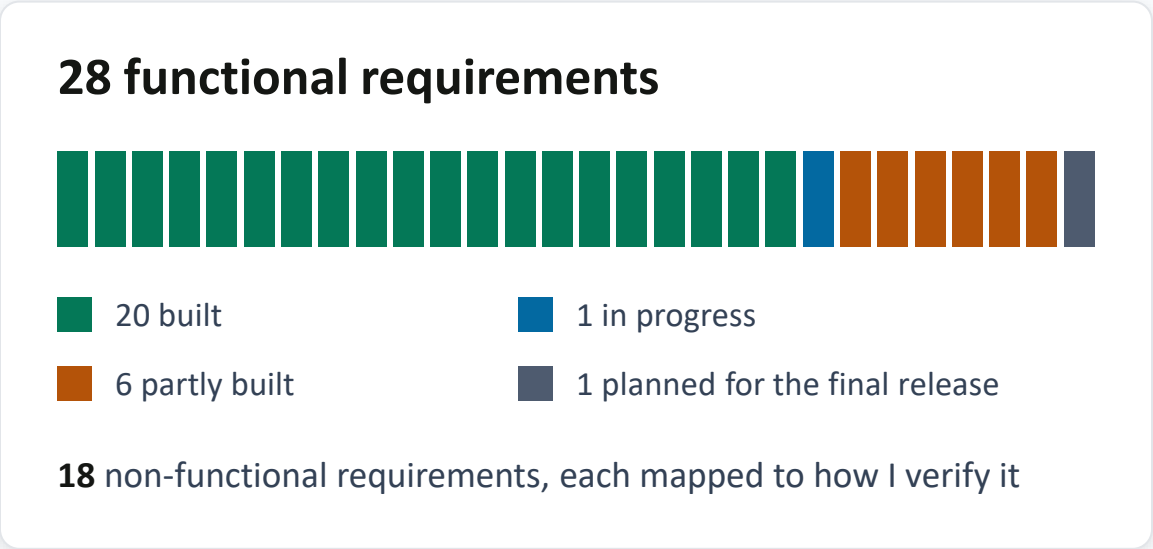
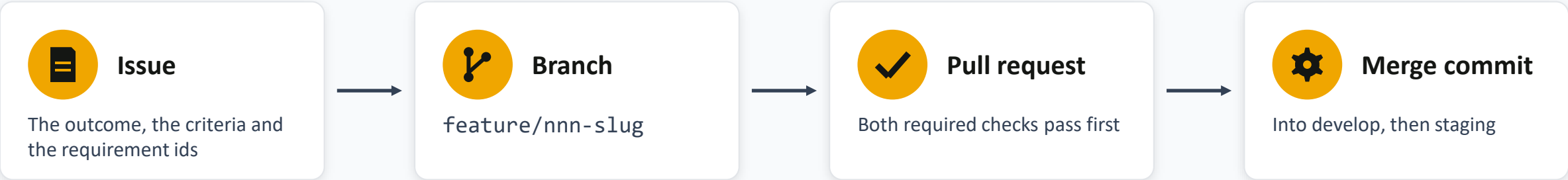
screens run on the API

01	02	03	04	05	06
07	08	09	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24

 Customer and counter, on the API

 Admin, still sample data, and it says so

Requirements and how they are traced

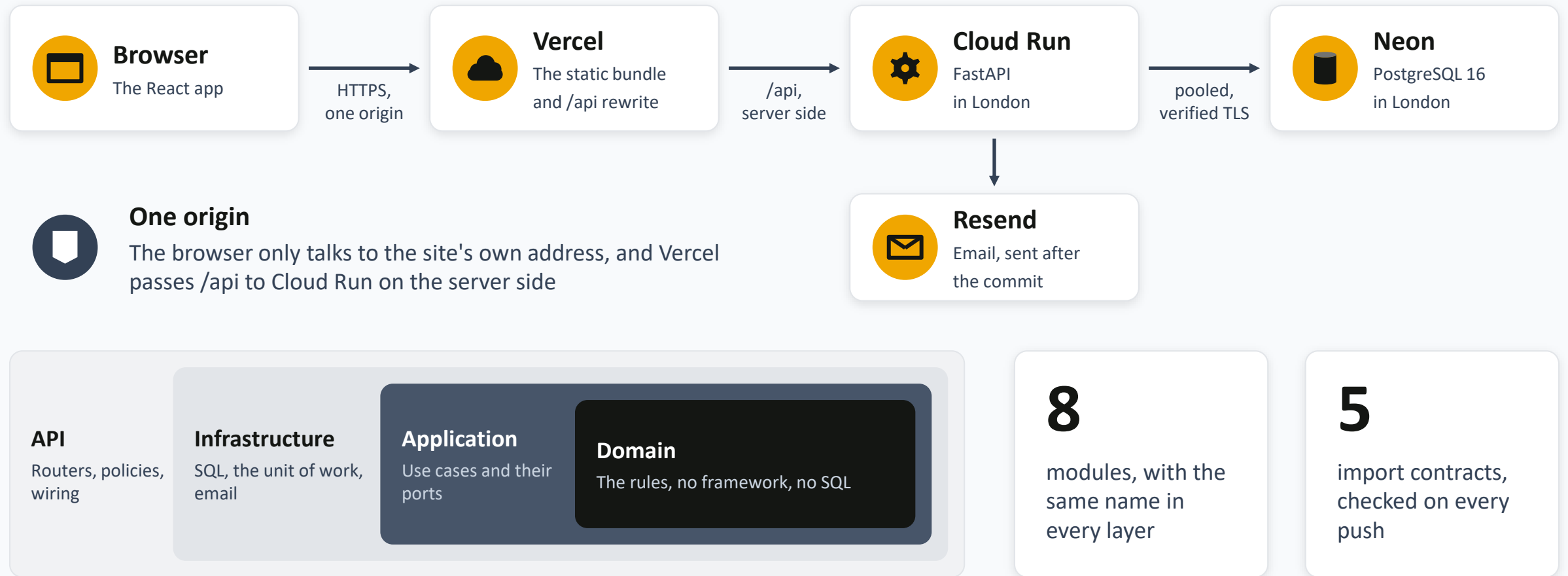


FR-07, followed through

No unit ever holds two overlapping bookings

Status	Built
Pull requests	#24 on branch 002, #38 on branch 005
Tests	test_exclusion_constraint.py test_concurrent_holds.py

Architecture

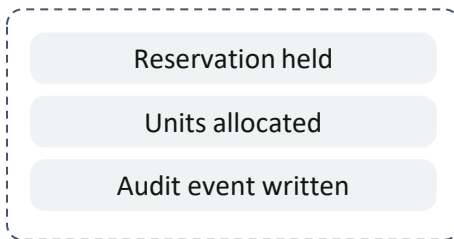


The four design patterns



Repository with Unit of Work

One transaction for a hold, its allocations and its audit event.

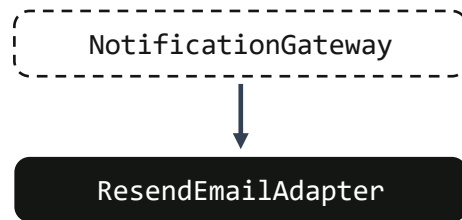


application/unit_of_work.py



Adapter

The email provider sits behind a port, with a transactional outbox.

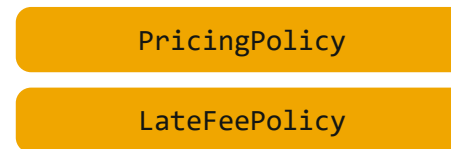


infrastructure/notification



Strategy

The pricing policy and the late fee policy, each in one place.



Standard, and fixed for tests

domain/policies



State

Eight reservation states. A move is refused unless the state allows it.



domain/states

The database

17

tables with singular names

6

hand written migrations



Keys and indexes

Composite keys and partial indexes, and tests that prove each index is used



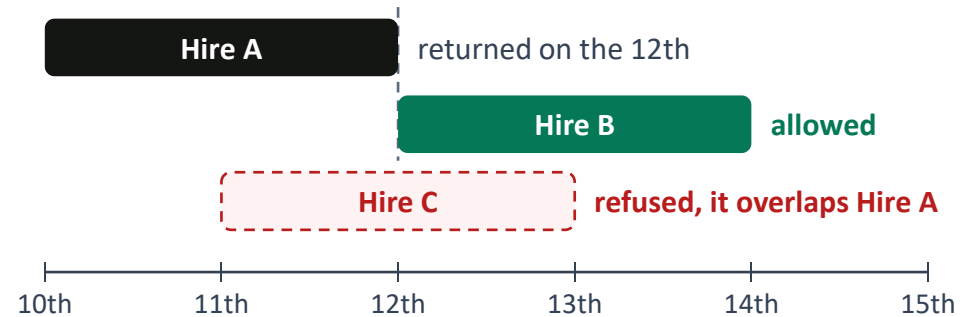
Two database roles

The running application cannot change the schema or rewrite the audit trail

THE CONSTRAINT

```
ALTER TABLE asset_allocation
ADD CONSTRAINT asset_allocation_no_overlap
EXCLUDE USING gist (
    asset_id WITH =,
    daterange(start_date, end_date, '[')') WITH &&
) WHERE (released_at IS NULL);
```

One unit, half open periods



Security



Sessions

A fifteen minute token in memory and a rotating refresh cookie



Deny by default

Ownership answered as not found, branch scope for counter staff

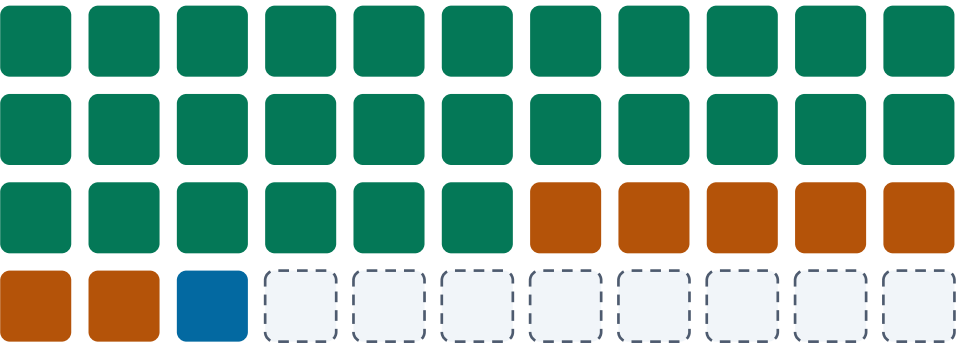


Lockout and throttles

Security headers on every answer, and no secret in the repository

44 controls from the design

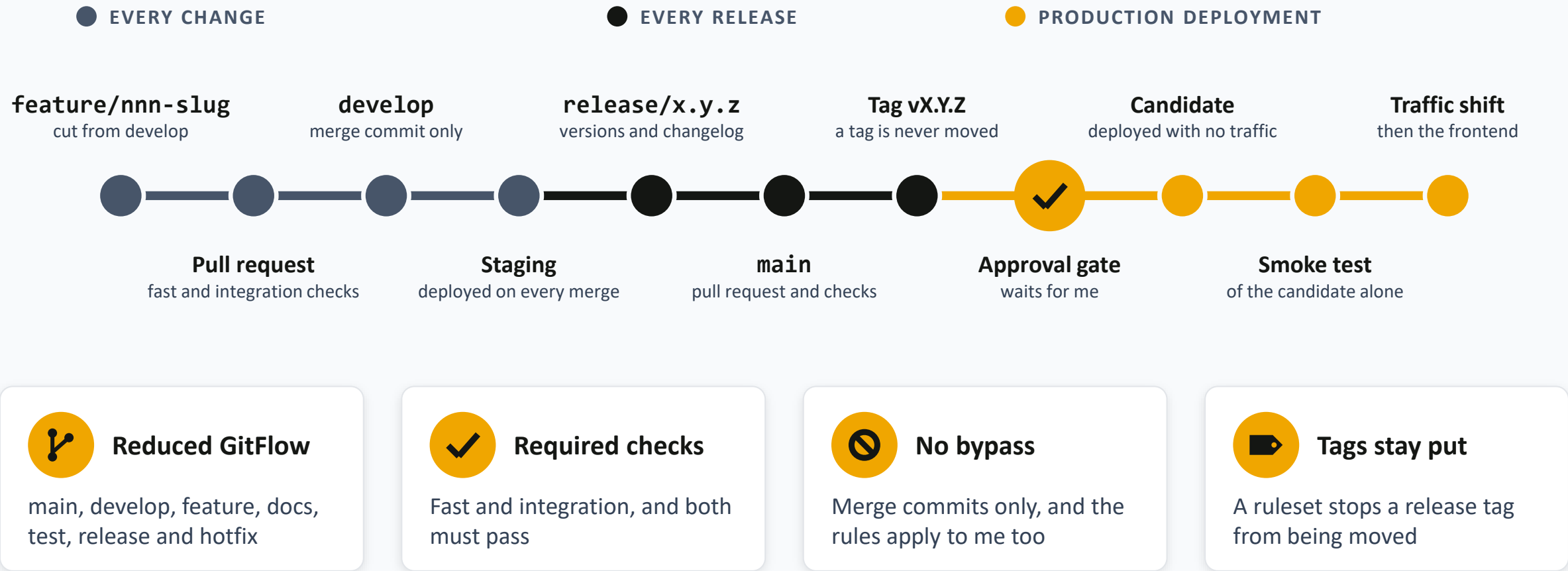
Each with its status and the file, test or step that shows it



- Green 28 in place
- Orange 7 partly in place
- Blue 1 replaced
- Dashed 8 planned

Still open: the photograph upload, the reporting export, alerting and the incident procedure

Pipeline and branching, a live look at GitHub



Hosting and the release procedure



Three hosts

Vercel for the site, Cloud Run in London for the API, Neon in London for the database



Keyless deployment

Workload Identity Federation, and every secret kept in Secret Manager



My approval, then a candidate

A release tag waits for my approval, then a candidate is smoke tested before it takes any traffic



One instance at most

With a budget alert, because this runs on a free trial

	Staging	Production
Deployed by	A merge into develop	A version tag on main, after my approval
Database	Neon branch staging	Neon branch main
New revision	Takes traffic at once	No traffic until its smoke test passes
Instances	At most one	At most one, while it runs as a demonstration



The API scales to zero, so the first request after a quiet spell waits for it to wake

Live demonstration

- 1

The customer journey on production

Search, hold, confirm and cancel, about one and a half minutes
- 2

The counter journey

A walk-in, checkout, the return and the deposit, about two minutes
- 3

The worked example from Task 1

Rental TSH-H-26-000098, opened on the return screen
- 4

The owner's view

The admin area and a branch dashboard, about half a minute

THE WORKED EXAMPLE, ON PRODUCTION

TSH-H-26-000098

SETTLED

Deposit held

R1,200.00

Late fee, two days at R120.00

R240.00

Withheld from the deposit, R208.70 plus R31.30 VAT

Released to the customer

R960.00

Due at the counter

R0.00

A test follows these figures through the API on every pull request.

Testing

3,431

backend tests

100%

of lines in the domain and application layers

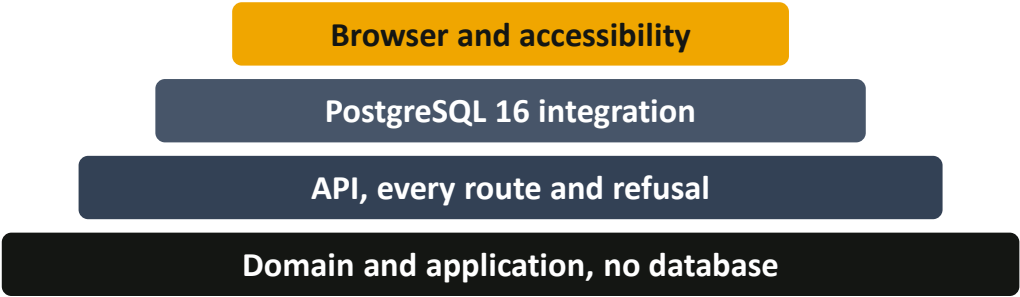
1,056

frontend unit and component tests

114

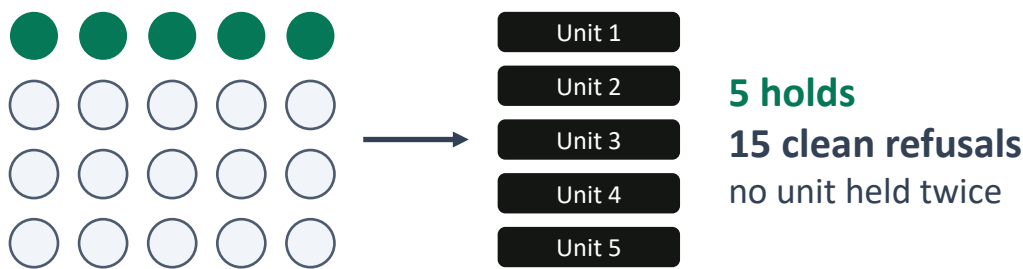
browser and accessibility tests on the real API

Four kinds of test



Coverage floor of 70 percent, mypy strict on 235 files, 5 layer import contracts

Twenty holds at once for five units



On every pull request, through HTTP, against PostgreSQL 16



Deviations from the design

THE DESIGN

Production capped at four instances

Email delivered to every customer

Checks every five and thirty minutes

A booking may start today at any hour

WHAT I BUILT

→ **One instance in both environments**

→ **Restricted mode, delivered to my own address only**

→ **No scheduled checks, and a smoke test on every deployment**

→ **Bookings for today stop at closing**

WHY

An instance cap is the only hard limit on cost while this runs on a free trial.

There is no verified sending domain, and a stranger should never get mail from a university project.

Checks every few minutes would keep the API and the database awake all day.

After closing nobody can collect, and the no show sweep would strike the customer at once.



What is left for Task 3

15 of 22 planned branches merged

v0.1.0 live

v0.2.0 live

v0.3.0 live

v1.0.0 in Task 3

001

002

003

004

005

006

007

008

009

010

011

012

013

014

015

016

017

018

019

020

021

022



Measurements and drills

Load, volume and cold start, the restore drill and the usability sessions



Alerting and the incident procedure

Alerts on token reuse and lockouts, and a written, rehearsed procedure



A customer's own data

Statement download, data export, retention period



Release v1.0.0

With the report and the user guide, once the system matches the whole design



Thank you



toolshed-hire.vercel.app



github.com/teon-artilionAI/toolshed-hire



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The README is the place to start

